

CE 310

Bootstrap Confidence Intervals

Week 14 · Session 1 · Quantifying Uncertainty Without Distributional Assumptions

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Defend This Number at a Public Hearing

60 seconds · pick A, B, or C

A city wants to drop a posted limit from 40 to 35 mph. Standard practice sets the limit from the **85th-percentile speed** in a spot study.

Your study: **120 vehicles**, mean 34.2 mph, SD 3.9 mph, **85th percentile = 38.5 mph**.

A council member asks the only question that matters: "**How sure are you about 38.5?**"

- **A)** Report $38.5 \pm t \cdot s / \sqrt{n}$ — the interval you have used since Week 6
- **B)** You can't give one — no standard-error formula exists for a percentile
- **C)** You can, but only by resampling the 120 speeds you already collected

✓ Answer: C — Let the Data Make Its Own Interval

Resample your 120 speeds, recompute the percentile, repeat

Draw 120 speeds **with replacement**, take the 85th percentile, repeat 1,000 times:

	Result
85th percentile of the study	38.5 mph
Bootstrap 95% confidence interval	36.6 – 39.5 mph
Resamples landing at or above 40 mph	0.2%

The last row is the sentence for the hearing: the data are very hard to reconcile with 40.

💡 No formula produced that interval — the data did. Any statistic you can compute, you can put an interval around.

Bootstrap vs. Monte Carlo

Both methods use repeated sampling to quantify uncertainty — but from different sources.

Week 13 — Monte Carlo

- Samples from *assumed* distributions
- Normal(0, 2°F), Normal(0.12, 0.01)
- Asks: if inputs vary like this, what does the output distribution look like?
- Requires: a model and distributional assumptions

Week 14 — Bootstrap

- Resamples from *your observed data*
- No distributional assumption
- Asks: how uncertain is the statistic I computed?
- Requires: only the data itself

Bootstrap answers a different question: not "what if?" but "how reliable is this estimate?"

The Bootstrap Algorithm

Three steps:

1. Start with your n observations — the original sample
2. Draw n observations **with replacement** → a bootstrap resample (same n , but some rows appear 2–3×, others not at all)
3. Compute the statistic of interest on this resample → record one bootstrap estimate

Repeat steps 2–3 $N_BOOT = 1,000$ times.

The distribution of 1,000 bootstrap estimates is the **bootstrap sampling distribution**. The 2.5th and 97.5th percentiles form the **95% bootstrap CI**.

Drawing n items with replacement from n items picks up only about **63 %** of them — $1 - 1/e$. Every resample leaves roughly a third of your data on the table, and that is exactly where the variation comes from.

Why Resampling Works at All

The bootstrap looks like cheating. You want to know how much your estimate would vary **across different samples** — but you have one sample, and cannot go and collect ninety-nine more. The move that makes it legitimate:

Your sample is the best picture of the population you have. So treat it as the population, and draw new samples from it.

What you wish you could do	What the bootstrap does
Draw 1,000 fresh samples of n from the population	Draw 1,000 resamples of n from your sample
Recompute the statistic each time	Recompute the statistic each time
See how much it varied	See how much it varied

That substitution has a name — the **plug-in principle**: wherever the theory asks for the population, plug in the sample.

 Everything the bootstrap can do rests on that one substitution — so a bootstrap is only as good as your sample is a picture of the population. A biased sample, or a very small one, breaks it at the root.

Quick Check — What Is a Resample Standing In For?

45 seconds · say it in one sentence

Your spot study caught **120 vehicles** on a Tuesday morning. You resample those 120 a thousand times and build an interval for the 85th-percentile speed.

- **A)** Each resample is a simulated *new spot study* of 120 vehicles
- **B)** Each resample is just a sample of the 120 drivers you already measured
- **C)** Both — and the gap between them is the assumption you are making

✓ C — and the Second Is Standing In for the First

Which is also where the method quietly fails

Mechanically it is **B**: you are drawing rows from a spreadsheet of 120 speeds. What you are *claiming* is **A**: that this behaves like going out and running the study again.

That claim holds only if the 120 vehicles genuinely resemble the traffic the limit will govern.

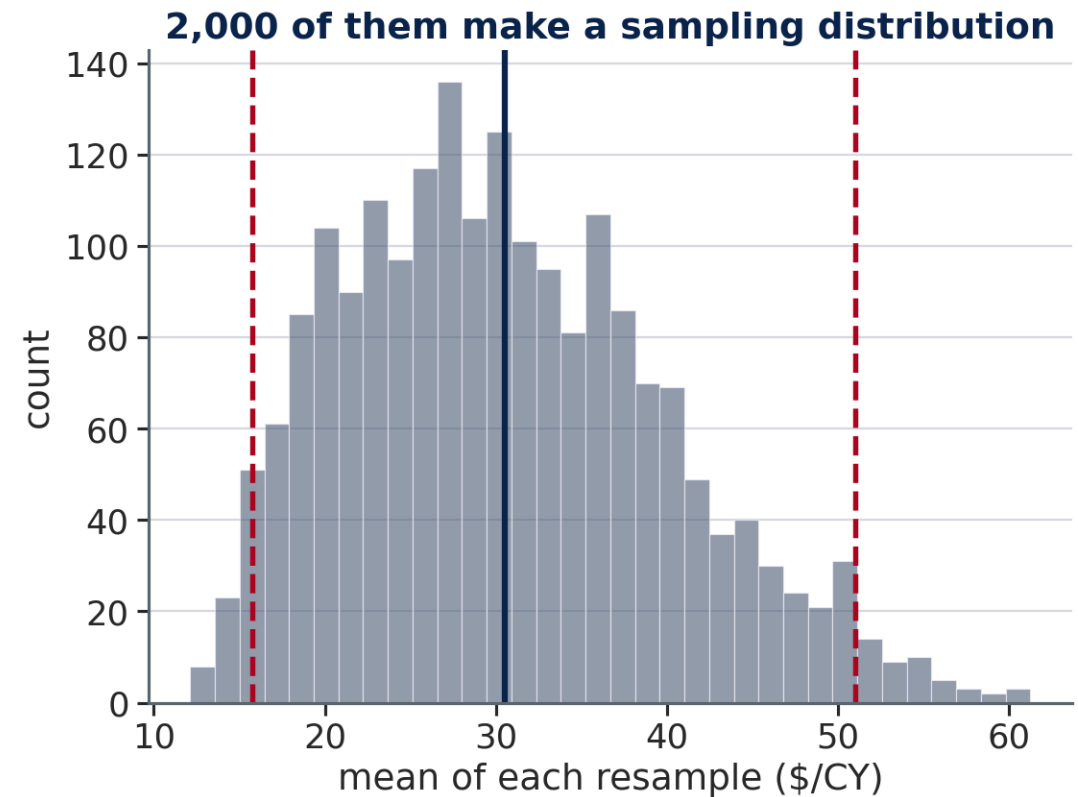
⚠ Tuesday morning is not Friday evening. If the study caught a commute peak and the posted limit governs all week, no amount of resampling repairs it — the bootstrap tells you what a *repeat of this study* would give, never what a *better* study would.

💡 Resampling quantifies **sampling variability**. It does nothing whatever about **bias**. A thousand resamples of a badly chosen 120 vehicles produce a beautifully tight interval around the wrong number.

The Mechanism, on Eight Rows

Draw n prices with replacement — gold = a repeat

original sample	12	19	8	45	22	15	96	27	mean 30.5
resample 1	19	22	12	27	96	15	8	15	mean 26.8
resample 2	12	8	27	19	45	45	15	15	mean 23.2
resample 3	12	45	96	12	8	8	22	45	mean 31.0



Each resample is the same size as the original but not the same contents — some rows arrive twice, others not at all. That reshuffling is the entire source of the variation.

A Bootstrap on Eight Numbers

The same eight rows as the diagram, now with numbers on them. Eight awarded unit prices, in dollars per CY:

12 · 15 · 18 · 19 · 22 · 26 · 31 · 45 — sample mean **\$23.50**

Draw eight **with replacement**, three times over:

Resample	The eight values drawn	Its mean
1	15, 15, 19, 22, 26, 26, 31, 45	\$24.88
2	12, 12, 18, 18, 19, 22, 26, 31	\$19.75
3	12, 18, 19, 22, 26, 31, 45, 45	\$27.25

Three resamples, three different means — and **\$23.50 never came back**. That scatter is the entire method.

💡 Do it a thousand times, sort the thousand means, and read off the 25th and the 975th. Those two numbers are your 95% confidence interval. There is no formula anywhere in that sentence.

Quick Check — Which of These Cannot Happen?

45 seconds · one is impossible, one is only surprising

Your sample, once more: **12 · 15 · 18 · 19 · 22 · 26 · 31 · 45**

Three candidate resamples:

- **A)** 12, 15, 15, 19, 26, 31, 45, 45
- **B)** 12, 18, 20, 22, 26, 31, 31, 45
- **C)** 45, 45, 45, 45, 45, 45, 45, 45

Which one could a bootstrap never produce — and is the other odd-looking one legal?

✓ B Is Impossible · C Is Perfectly Legal

A resample holds only values the sample already held

B contains a 20, and there is no 20 in your sample. A resample draws *rows you have*. It cannot invent a value sitting between two of them.

C is legal — the \$45 row drawn eight times running. Its probability is about **1 in 16.8 million**, so you will not see it, but nothing rules it out, and milder versions of it turn up constantly across a thousand resamples.

⚠ B is why a bootstrap cannot interpolate, and C is why it cannot exceed. Hold on to both — together they are the reason the flood maximum later in this lecture comes out with no usable interval at all.

Two Histograms That Get Confused

⚠ The histogram of your **data** and the histogram of your **bootstrap estimates** are different objects with different widths. Confusing them is the most common bootstrap error there is.

	Histogram of the data	Histogram of 1,000 bootstrap means
One bar counts	one observation	one whole resample's statistic
Its spread is	variability between jobs	uncertainty in your estimate
For awarded prices	SD $\approx$ \$33/CY, skewness 4.84	SD $\approx$ \$1.4/CY, near-Normal
It narrows when	never — it is what the world is	n grows

The data spread describes the world. The bootstrap spread describes **how well you have pinned down one number about the world** — and only the second one shrinks as you collect more data.

How Many Resamples Do You Need?

Week 13 asked this about simulation runs, and the answer has the same shape here — **it depends which part of the distribution you are reading.**

What you want out	Resamples needed
A bootstrap standard error	a few hundred is plenty
A 95% interval — the 2.5th and 97.5th percentiles	1,000 is the usual working minimum
A 99% interval , or a p-value below 0.01	10,000 and up

This course uses **N_BOOT = 1,000** throughout, which is standard practice for a 95% interval.

💡 There is a floor you cannot argue past. A permutation p-value from 1,000 shuffles can never come out smaller than $1/1,000$ — that is why the winner-discount test reported $p = 0.000$ rather than a smaller number. The resolution of the answer is set by how many times you ran it.

Bootstrap CI for a Mean — Validation

For a mean there *is* an analytical formula. Running the bootstrap on the same statistic tells you whether the machinery works before you point it at something no formula covers.

The 555 awarded roadway-excavation prices, mean **\$28.14/CY**:

95% CI for the mean awarded price	Interval	Width
Analytical t -interval	25.341 – 30.936	5.595
Percentile bootstrap	25.551 – 30.910	5.359

Close enough to trust the code — and not identical. The t -interval reaches exactly 2.797 either side of the mean, because subtracting and adding one quantity is what the formula does. The bootstrap reaches **2.587 below and 2.772 above**.

💡 Nothing instructed the bootstrap to lean upward. These prices are right-skewed — mean \$28.14/CY against a median of \$19.00/CY — and the resampled distribution inherited that shape.

Quick Check — Bootstrap vs. Analytical CI

60 seconds · commit before you hear the answer

Awarded excavation prices are badly behaved. Mean **\$28.14/CY**, median **\$19.00/CY**, skewness **4.84**, and a long tail running past \$200/CY. Nothing about this sample is Normal.

The t -interval formula assumes normality. So:

- **A)** The two intervals for the **mean** will agree closely anyway
- **B)** They will disagree badly — the data are far too skewed for the t -formula

✓ Answer: A — Skewed Data, Well-Behaved Mean

The assumption is about the estimator, not the sample

Analytical t -interval: (25.341, 30.936), width 5.595

Percentile bootstrap: (25.551, 30.910), width 5.359

A 4% difference in width, from a sample with skewness 4.84.

The t -formula never assumed the prices were Normal. It assumed the **sampling distribution of the mean** was approximately Normal — and at $n = 555$ the Central Limit Theorem supplies that, however ugly the underlying data.

💡 The CLT protects the **mean**. It does not protect a percentile, a slope, an R^2 , or a ratio. Every one of those is coming up in the next twenty minutes, and none of them gets this guarantee.

Bootstrap CI for Complex Statistics

No closed-form sampling distribution exists for a percentile, an R^2 , or a ratio. The bootstrap treats all of them exactly as it treated the mean.

Example: the 95th percentile of awarded price

An estimator setting a not-to-exceed line for roadway excavation cares about the upper tail. Five per cent of awarded jobs came in above **\$94.30/CY** — so that is the line.

	Value
Observed P95	\$94.30/CY
Bootstrap 95% CI	\$66.50 – \$106.50/CY
Interval width	\$40.00/CY — 42% of the estimate

💡 The mean was pinned to $\pm 10\%$; this is $\pm 21\%$. A tail is estimated from the few observations that reach it. Without the bootstrap it would carry no interval at all — not a narrow one, none.

Percentile Is Not the Only Interval

Every interval so far has been a **percentile bootstrap** — take the 2.5th and 97.5th percentiles of the resampled estimates and call that the 95% CI. It is the simplest choice and, for these statistics, a defensible one. It is not the only one.

Method	What it does	When you want it
Percentile	the 2.5th and 97.5th of the resamples	the default; fine when the bootstrap distribution is near-symmetric
Basic (reverse percentile)	reflects the resamples about the observed estimate	when the estimate is visibly biased
BCa	corrects for bias and skewness	small n , skewed statistics — the safest general choice

💡 Not something to memorise. The point is that "**bootstrap CI**" is **not one thing** — a report saying only "bootstrap 95% CI" has omitted which of these it used, and for a skewed statistic the three disagree.

Pairs Bootstrap for Regression Slopes

Resample whole **rows** — each row's (`logq` , `logp` , `L0W`) travels together. Resampling the predictor and the response independently would destroy the very relationship the regression exists to measure. Draw N rows with replacement, refit, record the slope, repeat 1,000 times.

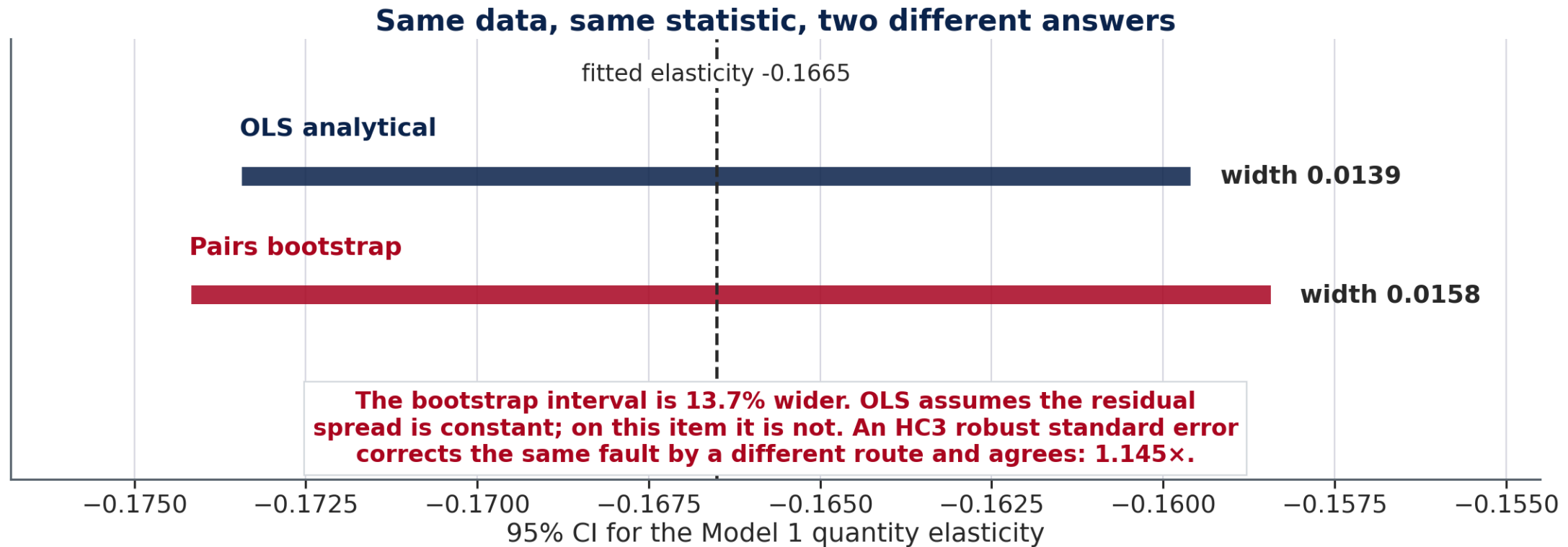
The alternative, a **residual bootstrap**, holds the predictors fixed and resamples only the residuals. That suits a designed experiment, where x was set at chosen levels. Bid data is **observed** — nobody assigned the job quantities — so resampling pairs is the honest choice, because it carries the variability in x through too.

The Model 1 quantity elasticity, the number every Week 12 forecast rested on:

95% CI for the elasticity	Range	Width
OLS analytical	−0.1735 – −0.1596	0.0139
Pairs bootstrap	−0.1742 – −0.1584	0.0158

The bootstrap interval is **13.7% wider**. You compute that ratio yourself as `ANSWER_B8`.

The Two Intervals, Side by Side



Same 4,369 rows, same statistic, two answers. The bootstrap is wider because it does not assume something about this data that happens to be false.

Permutation Test — Assumption-Free Hypothesis Testing

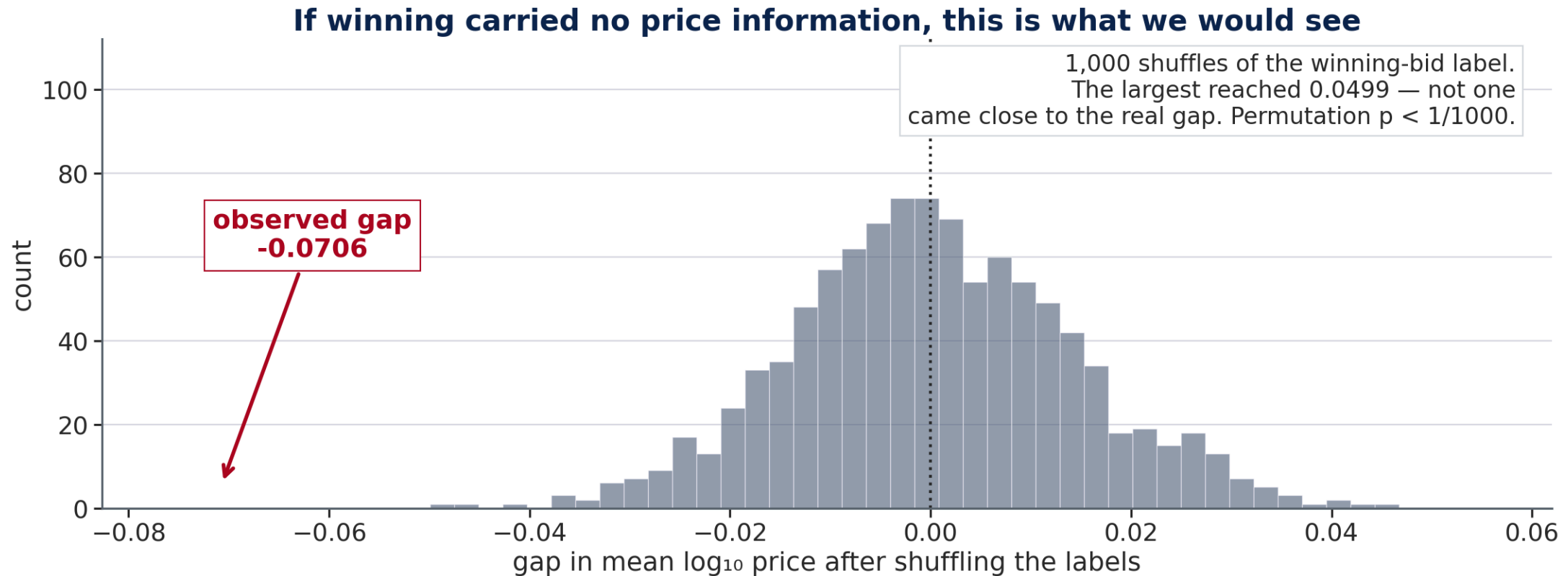
A permutation test asks: if the group labels carried no information, how often would a shuffle produce a gap as large as the one we actually saw?

The winner discount, Week 11 revisited:

1. Observed: mean `logp` for winners minus losers = **-0.0706** — winners price about **15% below** losers
2. Pool all 4,369 bids; shuffle the `LOW` labels at random
3. Recompute the gap on the shuffled labels
4. Repeat 1,000 times → how many shuffles reach 0.0706?

None. The largest of the thousand reached **0.0499**. The permutation p-value is reported as 0.000, which means $p < 1/1000$, and the Week 11 t -test agrees at 1.65×10^{-7} .

The Permutation Null



A thousand shuffles of the winning-bid label, and the gap never leaves ± 0.05 . Ours was -0.0706 . The null distribution was built from the data, not assumed.

Quick Check — Is the Permutation Test Valid Here?

60 seconds · one subtlety, worth catching

The 4,369 bids are **not** 4,369 independent observations. They come from **893 projects**, about **4.9 bidders each** — and every bidder on a project is pricing the same site, the same plans, the same season.

The test we just ran shuffled the **LOW** label freely across all 4,369 rows.

- **A)** Invalid — a permutation test needs independent observations, and these are clustered
- **B)** Valid as written — shuffling labels assumes nothing, so clustering cannot hurt it
- **C)** Valid in principle, but the shuffle should respect the design: one winner per project

✓ Answer: C — Match the Shuffle to the Design

Exchangeability is the assumption — not independence

A permutation test's only requirement is that **under H_0 the labels are exchangeable** — a claim about labels, not about rows being independent draws.

But a free global shuffle builds null datasets no bid letting could produce: four winners on one project, none on another. Permuting **within each project** respects how the work was let.

Shuffle	Null SD	Observed gap, in null SDs
Global, all 4,369 rows	0.0134	5.3
Within project	0.0090	7.8

💡 Conditioning on the project removes between-project price variation, which was never relevant to this comparison. The stricter test is the *more* significant one. Finding a flaw in a method does not mean the answer flips.

Bootstrap or Permutation — Which Question Are You Asking?

Both methods resample. They answer **different questions**, and choosing the wrong one gives a technically correct answer to something nobody asked.

Bootstrap

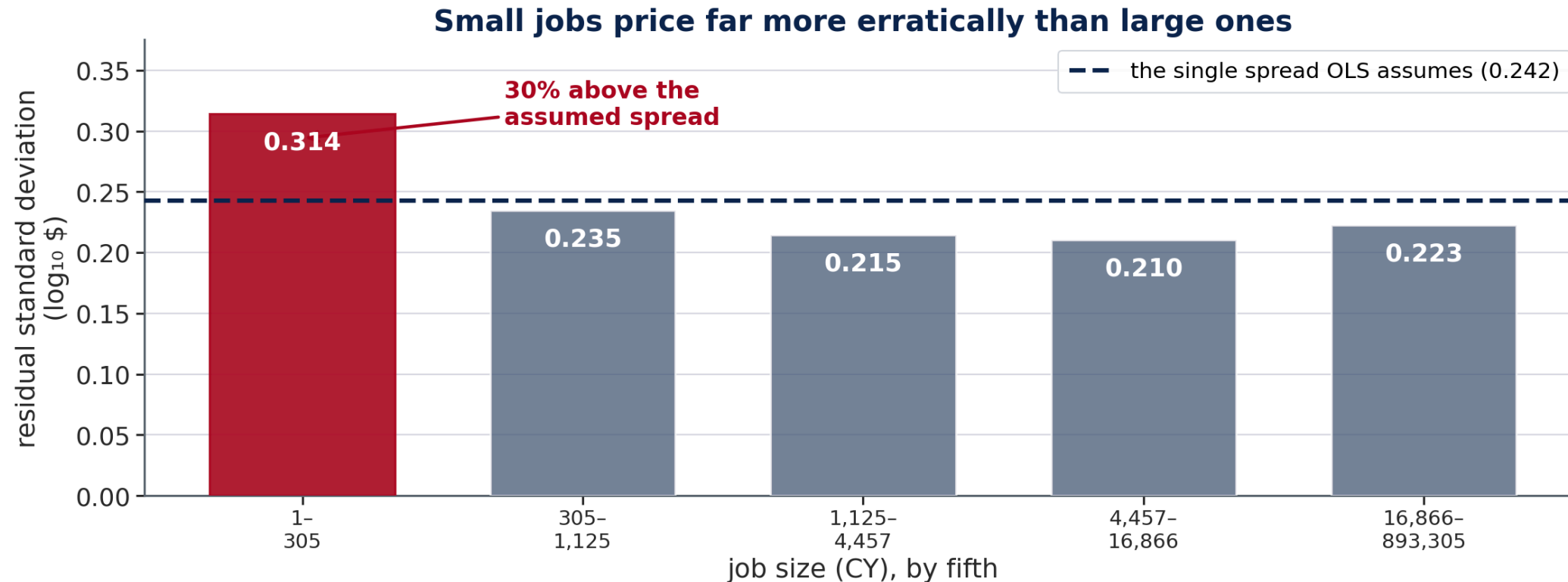
- Resamples **with replacement**, labels left alone
- Asks: *how uncertain is this estimate?*
- Produces: a confidence interval
- "The elasticity is -0.166 , and it lies in -0.174 to -0.158 "

Permutation

- **Shuffles the labels**, data left alone
- Asks: *could a gap this big be chance?*
- Produces: a p-value
- "No shuffle in a thousand reached our gap"

💡 They are not rivals. The interval says **how big** the effect is; the p-value says only whether it is distinguishable from nothing. A p-value with no interval tells you the effect is real without telling you whether it matters.

When Bootstrap and Analytical CIs Diverge



OLS earns its standard error by assuming one spread describes every job. Small jobs miss the fitted line by **0.314** in $\log_{10}$ dollars; the rest sit near **0.21–0.24**.

Quick Check — Bootstrap the Largest Flood on Record

60 seconds · this one has a trap

You hold **80 years** of annual peak discharge for a river. The largest in the record is **41,200 cfs**.

You want an interval for the **maximum** — the biggest flow this river produces — so you resample the 80 years with replacement 1,000 times and take the largest value each time.

What does the bootstrap distribution of that maximum look like?

- **A)** Roughly symmetric about 41,200 cfs
- **B)** It never exceeds 41,200, and most resamples land exactly on it
- **C)** A little wider than the data, in both directions

✓ B — The Bootstrap Cannot Invent a Bigger Flood

The one statistic resampling genuinely cannot estimate

A resample is drawn **from the 80 years you have**, so it cannot contain a value the record does not. The bootstrap maximum is **capped at 41,200 cfs** — and since about 63% of resamples include the record year, roughly two-thirds land exactly on it.

The result is a spike at 41,200 with a one-sided tail below. **That is not a confidence interval.** It is an artefact of the method meeting a statistic it cannot handle.

⚠ What broke is the **plug-in principle**. Your sample stands in well for the population *in the middle*, where you have many observations. In the far tail it stands in badly — and a maximum lives entirely in the far tail.

💡 This is why hydrologists fit an extreme-value distribution instead. When you need the tail beyond your data, you need a **model of the tail's shape** — resampling can only reuse what you already saw.

Limitations of Bootstrap

The bootstrap removes an assumption about the *shape* of a sampling distribution. It does not remove the others.

- **A biased sample stays biased** — resampling 555 awarded Texas jobs describes awarded Texas jobs. No amount of resampling reaches the ones that were never let.
- **Very small n** — below about 20 the resamples are too coarse to trace a distribution
- **Extreme tails** — a 99.9th percentile rests on a couple of observations, and the bootstrap can only reuse those
- **Clustered rows** — the standard bootstrap treats rows as independent. These bids come about five to a project, sharing a site, a set of plans and a season.

A **cluster bootstrap** resamples whole projects instead of individual bids — the same idea as the block bootstrap for time series.

Reporting a Bootstrap Interval

A number without its provenance cannot be checked. When a bootstrap result goes into a memo, **five things travel with it**:

1. **The statistic** — the 95th percentile of awarded unit price
2. **The point estimate** — \$94.30/CY
3. **The interval and its level** — 95% CI, \$66.50 to \$106.50/CY
4. **The method** — percentile bootstrap; pairs resampling wherever a model was refit
5. **n and N_BOOT** — 555 observations, 1,000 resamples

"The 95th percentile of awarded unit price is \$94.30/CY (95% percentile-bootstrap CI: \$66.50 – \$106.50; $n = 555$, 1,000 resamples)."

💡 That sentence is the deliverable of the whole week: it states a number, admits what it does not know, and lets the reader reproduce it — the difference between an engineering estimate and a guess with a decimal point on it.

Bridge to Machine Learning — Ensemble Methods

Bootstrap concept	ML equivalent	Connection
Resample with replacement	Bagging (Bootstrap Aggregating)	Core of random forest training
Bootstrap CI for R^2	Model stability evaluation	How much does fit vary across folds?
Pairs bootstrap for slope	Cross-validation	Evaluate estimate stability on held-out data
Permutation test	Permutation feature importance	Shuffle one feature; measure accuracy drop

Random forest is bootstrap applied directly: each tree is trained on a bootstrap resample of the data. The ensemble prediction is the average across 100–1,000 trees, reducing variance compared to a single fit — exactly the same principle as averaging bootstrap resamples to estimate a statistic.

Three Deliverables This Week

 **No Friday in-class exercise this week.** Thanksgiving cancels Fri Nov 27. Monday and Wednesday are normal; you finish the exercise on your own time.

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 11:59 PM Sunday, Nov 29
In-Class Exercise	The Colab notebook — all sections, plots, and written answers	Start Wed, finish on your own · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

Office hours run as normal.

 Three submissions, three D2L Assignments folders. The quiz closes 11:59 PM Sunday, Nov 29; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 15: Capstone Project

Every tool from Weeks 9–14, applied to **a dataset your team chooses**:

- **EDA** (Weeks 2–4): describe the distribution, the groups, the relationship
- **Regression** (Weeks 9–10): Model 1 (X alone) and Model 2 (X + your group variable)
- **Hypothesis test** (Week 11): is the group difference statistically real?
- **Bootstrap** (Week 14): a CI on your regression slope, no formula required
- **Monte Carlo** (Week 13): the range your outcome could take under uncertainty

Your team picks one numeric predictor **X**, one numeric outcome **Y**, and one **grouping variable** that splits the data in two — from any real civil or architectural engineering source. Five ready-made building-energy datasets remain available if you would rather not go looking.

Week 16: a **4-minute client briefing** plus 1 minute of questions, then a cross-team debrief. No Python tutorial in Week 15 — the in-class exercise *is* the capstone.